

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO3_AT3 — VISUALIZATION CRITIQUE & REDESIGN

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO3 – Evaluation (BL5)	Assessment:	Assessment Tool 3 – Critique & Redesign
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO3 Statement:	<i>Evaluate flawed or misleading data visualizations, critique their specific shortcomings, and justify an improved redesign.(BL5)</i>		
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Section / Batch:	Slot C	Date:	18/08/2026

Code of Conduct

I Jothika M (192324235) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: *Jothika M*

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Each answer must go beyond simply naming a flaw: explain WHY it is misleading and what a reader could wrongly conclude from it.
- Your redesign must be a specific, justified chart proposal — not just “use a different chart.”
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – VISUALIZATION CRITIQUE & REDESIGN

Problem 2 (5 Marks)

Scenario: A quarterly sales report uses a 3D, tilted pie chart to compare 4 regions' revenue share.

- a) Critique the visual distortion introduced by the 3D tilt.
- b) Redesign: propose a flat alternative and explain why it fixes the distortion.

Problem 9 (5 Marks)

Scenario: A company's stacked bar chart of sales by product uses two nearly-identical shades of blue for two very different product categories.

- a) Critique this color choice.
- b) Redesign: propose an improved color scheme.

Problem 11 (5 Marks)

Scenario: A bar chart of city populations (ranging from 5,000 to 20,000,000) uses a linear y-axis, making all but the largest city's bar nearly invisible.

- a) Critique the specific problem this creates for the reader.
- b) Redesign: propose the correct axis type and justify it.

Problem 18 (5 Marks)

Scenario: A scatterplot of 100,000 transaction records is shown as a single dense black blob due to heavy overplotting, hiding any underlying pattern.

- a) Critique why this chart fails to communicate anything useful.
- b) Redesign: propose an alternative chart or technique that would handle this volume of data.

Problem 25 (5 Marks)

Scenario: A parallel sets diagram connecting 3 categorical variables uses the same color for several unrelated categories, making it hard to trace any single ribbon through the diagram.

- a) Critique this specific design flaw.
- b) Redesign: propose how you would fix the color scheme to make the flows traceable.

Rubrics for Calculation
ASSESSMENT RUBRIC
AT3 — Visualization Critique & Redesign

CO3 · BT5: Evaluate ·

This rubric is used to assess student responses to the CO3-AT3 Visualization Critique & Redesign problems, in which students are given a flawed or misleading chart, must critique its specific shortcomings, and propose a justified redesign.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the specific visualization principle being violated (e.g., proportional ink, axis scaling, color perception, overplotting) in the given flawed chart.	Good understanding of the relevant principle, with only minor gaps in identifying exactly what is violated.	Basic understanding, but with some misconceptions about which principle applies to the flaw shown.	Poor understanding of core principles; misidentifies the nature of the flaw entirely.
Explanation	25%	Clearly explains why the flaw is misleading, including what a reader could wrongly conclude as a result — not just that something is 'wrong.'	Explains the flaw with adequate reasoning, covering most of why it misleads a reader.	Identifies the flaw but explanation is generic or only loosely tied to its misleading effect.	Little or no explanation of why the flaw is misleading; the critique is a bare assertion.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Application	20%	Proposes a specific, well-justified redesign that directly resolves the identified flaw, correctly applying appropriate chart-selection and design principles.	Proposes a workable redesign that resolves the flaw, with only minor gaps in justification.	Proposes a redesign, but it only partially resolves the flaw or lacks sufficient justification.	Fails to propose a workable redesign, or the redesign does not address the identified flaw.
Organization	15%	The response is clearly structured, addressing both the critique and the redesign in a logical sequence with coherent overall flow.	The response is organized and addresses both parts, with only minor issues in flow or clarity.	Basic structure; both parts are addressed but the response lacks clarity or a logical sequence.	Poorly organized; the response is incoherent, and one or both parts are left unaddressed.
Accuracy	10%	Both the critique and the redesign are completely accurate and well-suited to the specific chart and data described.	Mostly accurate, with only minor omissions or a small error in the critique or redesign.	Partially correct; the critique or redesign contains notable inaccuracies or mismatches to the scenario.	Largely incorrect or inappropriate, with major gaps in both the critique and the redesign.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Problem 2 – 3D Tilted Pie Chart (5 Marks)

Introduction

A pie chart represents parts of a whole using circular slices. A 3D tilted pie chart adds perspective, but this perspective can distort the apparent size of the slices. In a quarterly sales report, this can lead to incorrect comparison of revenue shares.

Scenario

A quarterly sales report uses a **3D tilted pie chart** to compare the revenue share of four regions

a) Critique the Visual Distortion

The 3D tilt introduces **perspective distortion**. Slices located toward the front of the chart appear larger than equally sized slices at the back. Therefore, the visual appearance does not accurately represent the actual revenue proportions.

This can cause the reader to overestimate the revenue share of a region whose slice is positioned toward the front.

b) Redesign

A **flat 2D pie chart** should be used instead.

Removing the 3D perspective ensures that the **angle and area of each slice directly represent its actual revenue share**. A bar chart could also be used if precise comparison between regions is required.

Evidence

The 3D effect changes the apparent visual size of the slices even though their actual percentages remain unchanged.

Interpretation

The 3D chart may make the front region appear more important than a similarly sized region at the back. A flat 2D pie chart removes this misleading perspective.

R Code – 2D Pie Chart

```

R
library(ggplot2)

region <- data.frame(
  Region = c("North", "South", "East", "West"),
  Revenue = c(40, 30, 20, 10)
)

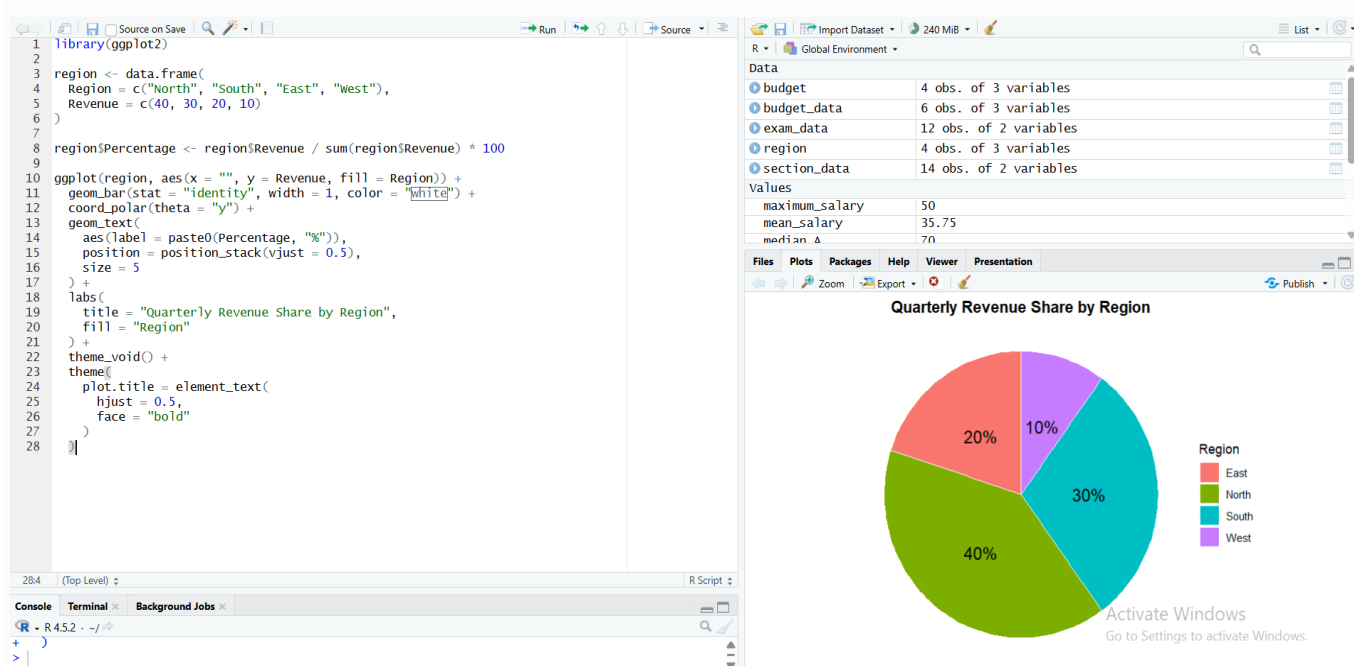
region$Percentage <- region$Revenue / sum(region$Revenue) * 100

ggplot(region, aes(x = "", y = Revenue, fill = Region)) +
  geom_bar(stat = "identity", width = 1, color = "white") +
  coord_polar(theta = "y") +
  geom_text(
    aes(label = paste0(Percentage, "%")),
    position = position_stack(vjust = 0.5),
    size = 5
  ) +
  labs(
    title = "Quarterly Revenue Share by Region",
    fill = "Region"
  ) +
  theme_void() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
)

```

Expected Result

The redesigned chart is a **flat 2D pie chart**, where each region's slice is proportional to its actual revenue share.



Conclusion

The 3D tilt creates perspective distortion and can make front slices appear larger than they actually are. A **flat 2D pie chart** provides a more accurate representation of revenue shares.

Problem 9 – Stacked Bar Chart Color Choice (5 Marks)

Introduction

Color is an important visual encoding in a stacked bar chart. Categories should have sufficiently different colors so that readers can easily distinguish them.

Scenario

A company's stacked bar chart uses **two nearly-identical shades of blue** for two very different product categories.

a) Critique the Color Choice

Using two nearly identical blue shades makes it difficult to distinguish the two product categories.

This is particularly problematic in a stacked bar chart because the two segments are directly adjacent to one another. The reader may not know where one product category ends and the other begins.

b) Redesign

Use **clearly distinct, high-contrast colors** for the product categories.

For example:

- Product A → Blue
- Product B → Orange

A legend or direct labels should also be included so that the chart does not depend only on color.

Evidence

Nearly identical colors make adjacent stacked segments difficult to separate visually.

Interpretation

Using contrasting colors makes it easier to identify each product and compare its contribution to total sales.

R Code – Improved Stacked Bar Chart

```
R
library(ggplot2)

sales <- data.frame(
  Year = c("2025", "2025", "2026", "2026"),
  Product = c(
    "Product A",
    "Product B",
    "Product A",
    "Product B"
  ),
  Sales = c(60, 40, 70, 30)
)

ggplot(
  sales,
  aes(x = Year, y = Sales, fill = Product)
) +
  geom_bar(
    stat = "identity",
    color = "white"
```

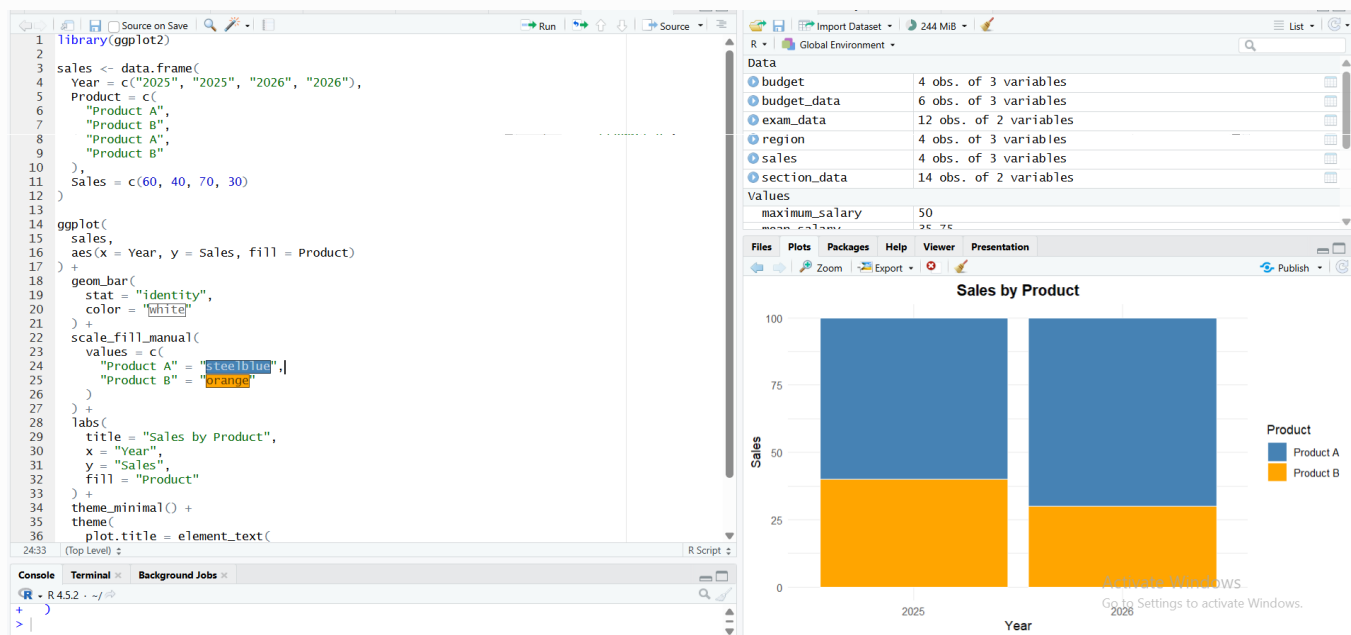
```

) +
scale_fill_manual(
  values = c(
    "Product A" = "steelblue",
    "Product B" = "orange"
  )
) +
labs(
  title = "Sales by Product",
  x = "Year",
  y = "Sales",
  fill = "Product"
) +
theme_minimal() +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold"
  )
)

```

Expected Result

The two product categories are clearly separated using contrasting colors.



Conclusion

The original color scheme reduces readability because the two blues are too similar. A **high-contrast color scheme with a clear legend** makes the stacked bar chart easier to understand.

Problem 11 – City Population with Linear Axis (5 Marks)

Introduction

An axis scale determines how numerical values are displayed. A linear scale works well when values have a relatively similar range. However, when values range from thousands to millions, a logarithmic scale can provide a clearer representation.

Scenario

A city population bar chart contains values from **5,000 to 20,000,000**, but uses a linear y-axis.

a) Critique the Specific Problem

The linear y-axis is dominated by the largest city.

Because the range is extremely large, the smaller cities have bars that are compressed close to zero. This makes it difficult or almost impossible for the reader to compare the smaller populations.

b) Redesign

Use a **logarithmic y-axis**.

A logarithmic scale represents values based on ratios rather than absolute differences. This keeps both small and large population values visually distinguishable.

Evidence

The population range is:

5,000 → 20,000,000

This covers several orders of magnitude, so a linear scale gives the largest city overwhelming visual dominance.

Interpretation

The smaller cities are not actually insignificant; they only appear insignificant because of the linear scale.

A logarithmic scale reveals the relative differences more effectively.

R Code – Logarithmic Y-Axis

```
R
library(ggplot2)

population <- data.frame(
  City = c(
    "City A",
    "City B",
    "City C",
    "City D",
    "City E"
  ),
  Population = c(
    5000,
    25000,
    150000,
    2000000,
    20000000
  )
)

ggplot(
  population,
  aes(x = City, y = Population)
) +
  geom_bar(
    stat = "identity",
    fill = "steelblue"
  ) +
  scale_y_log10() +
  labs(
    title = "City Population - Logarithmic Scale",
    x = "City",
    y = "Population (Log Scale)"
  ) +
```

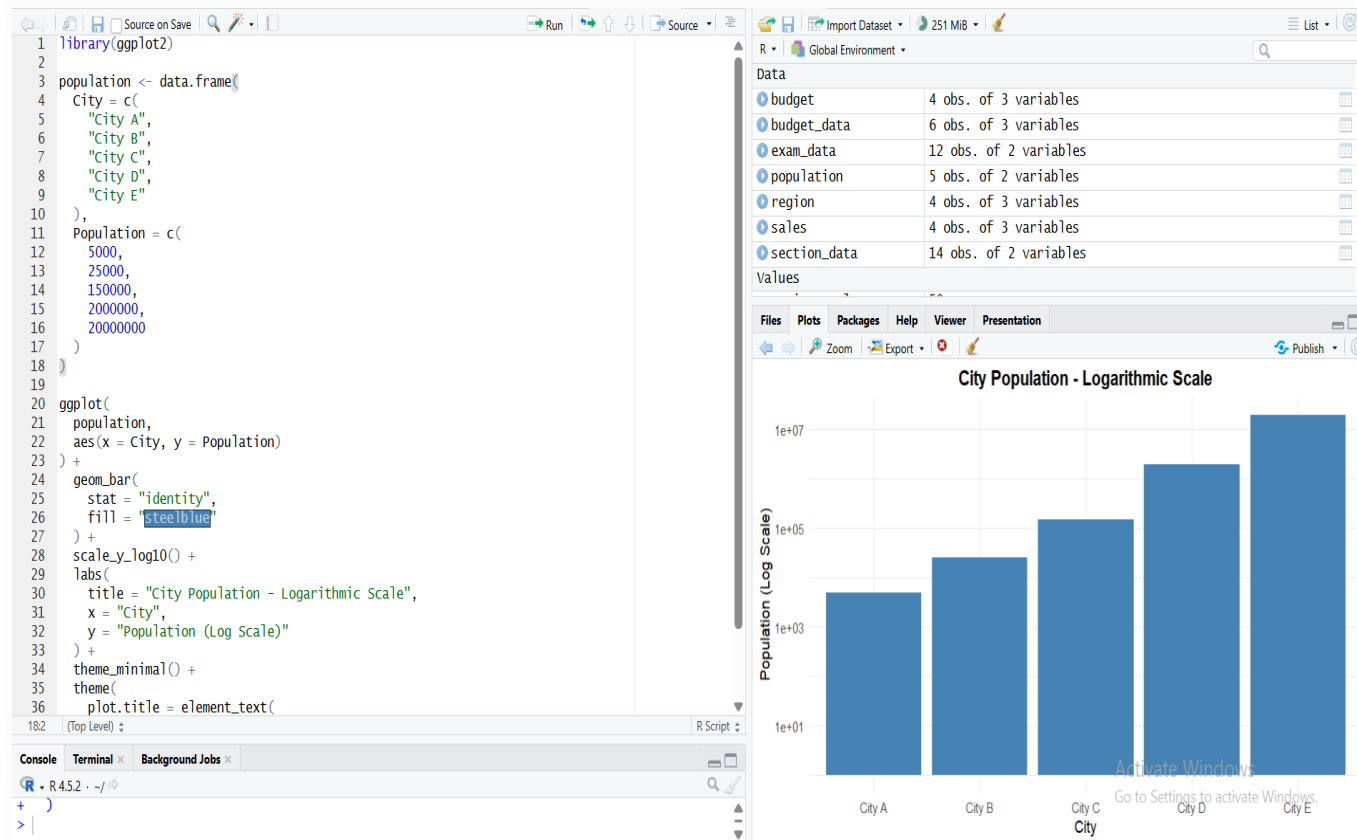
```

theme_minimal() +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold"
  )
)

```

Expected Result

All cities remain visible because the y-axis uses a logarithmic scale.



Conclusion

The linear scale makes smaller cities nearly invisible. A **logarithmic y-axis** is more suitable for population values covering a very wide range.

Problem 18 – Scatterplot with 100,000 Transactions (5 Marks)

Introduction

A scatterplot represents individual observations using points. It works well for moderate datasets, but very large datasets can create **overplotting**, where many points overlap.

Scenario

A scatterplot contains **100,000 transaction records** and appears as one dense black blob.

a) Critique Why the Chart Fails

The chart suffers from severe **overplotting**. Thousands of observations overlap in the same areas, causing individual points to merge into a dense black mass.

As a result, the reader cannot easily identify:

- Data concentration
- Relationships
- Clusters
- Density
- Underlying patterns

The raw scatterplot therefore fails to communicate the structure of the data effectively.

b) Redesign

Use a **2D density plot** or **hexbin plot**.

A hexbin plot groups nearby observations into cells and uses color intensity to indicate how many observations are present in each cell. This is much more suitable for very large datasets.

Evidence

With 100,000 points, many points occupy the same area of the graph. Therefore, plotting every observation individually produces excessive overlap.

Interpretation

The density of transactions is more important than seeing every individual point. A density-based visualization shows where transactions are concentrated.

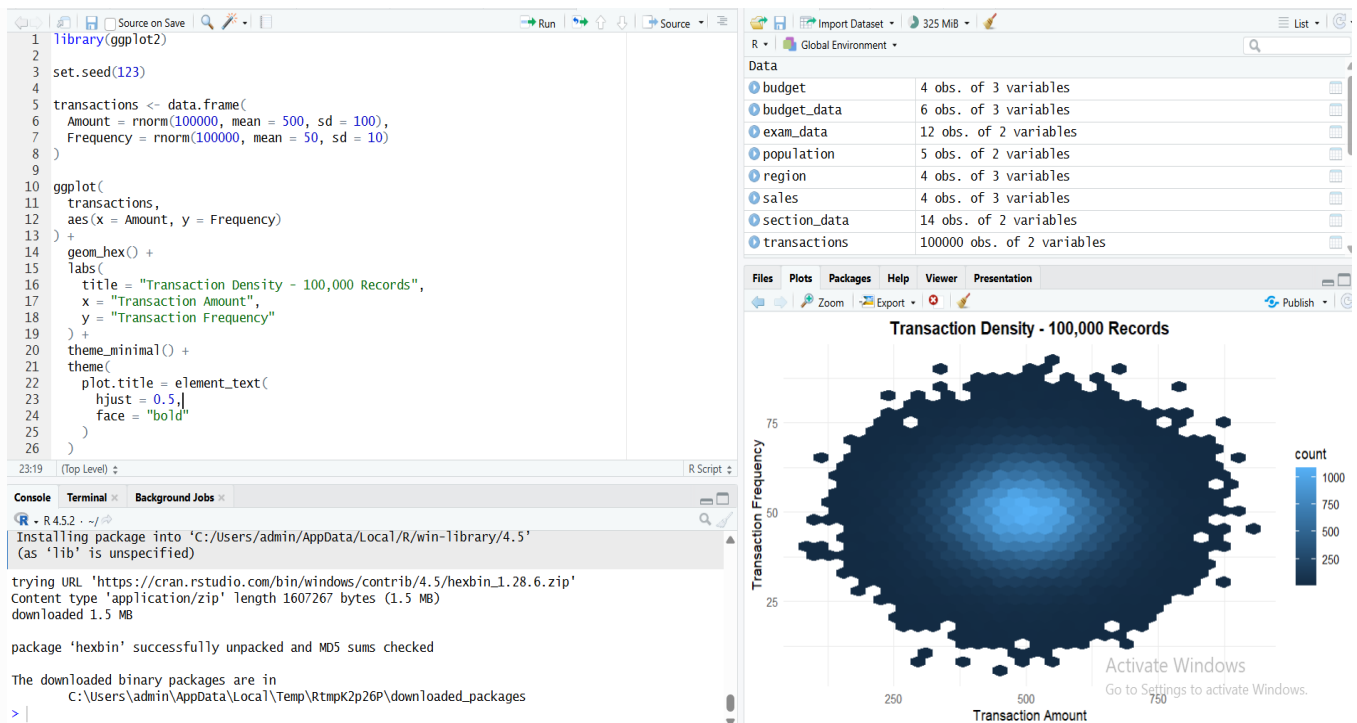
R Code – 2D Density Plot

```
R
library(ggplot2)

set.seed(123)

transactions <- data.frame(
  Amount = rnorm(100000, mean = 500, sd = 100),
  Frequency = rnorm(100000, mean = 50, sd = 10)
)

ggplot(
  transactions,
  aes(x = Amount, y = Frequency)
) +
  geom_hex() +
  labs(
    title = "Transaction Density - 100,000 Records",
    x = "Transaction Amount",
    y = "Transaction Frequency"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )
)
```



Alternative R Code – 2D Density

```

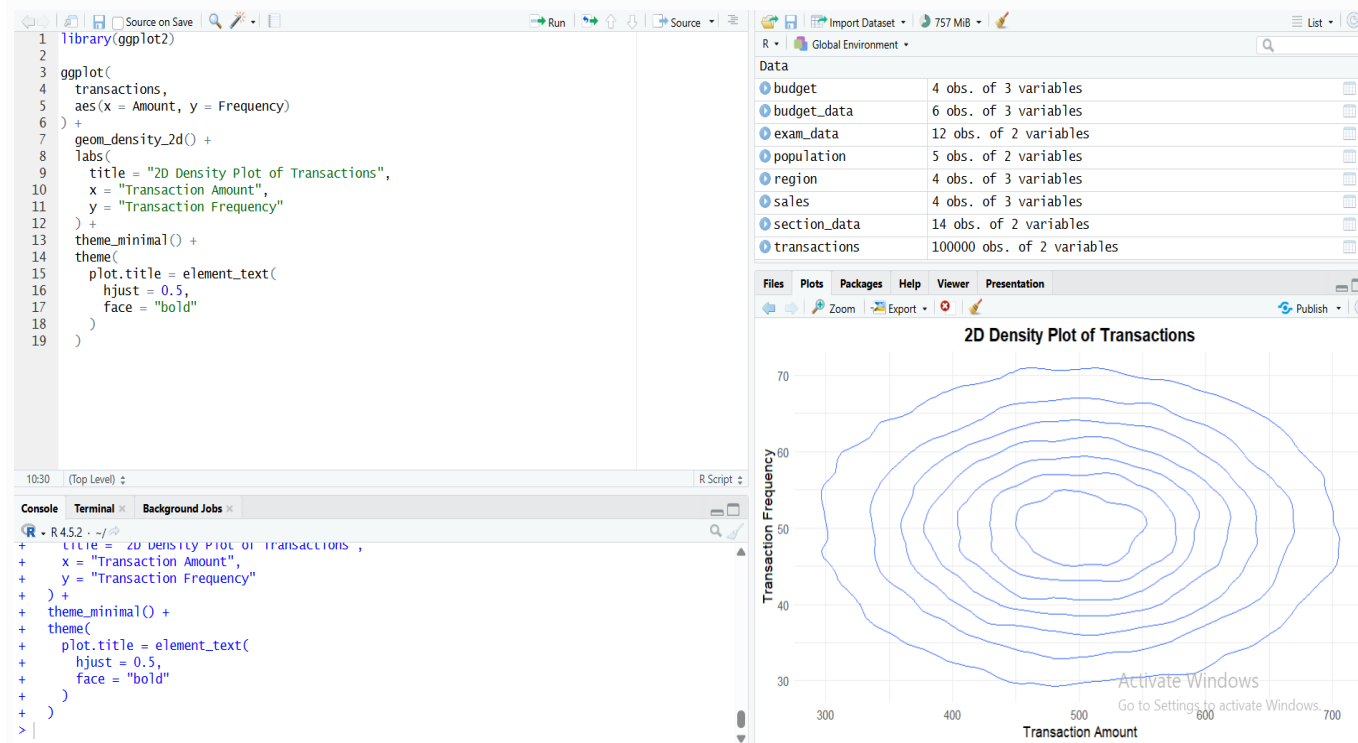
R
library(ggplot2)

ggplot(
  transactions,
  aes(x = Amount, y = Frequency)
) +
  geom_density_2d() +
  labs(
    title = "2D Density Plot of Transactions",
    x = "Transaction Amount",
    y = "Transaction Frequency"
  ) +
  theme_minimal() +
  theme(
    plot.title = element_text(
      hjust = 0.5,
      face = "bold"
    )
  )

```

Expected Result

Instead of one black blob, the plot shows areas where transaction observations are concentrated.



Conclusion

A normal scatterplot is unsuitable for 100,000 records because of overplotting. A **hexbin plot** or **2D density plot** provides a clearer representation of the underlying transaction pattern.

Problem 25 – Parallel Sets Diagram (5 Marks)

Introduction

A parallel sets diagram is used to visualize relationships and flows between multiple categorical variables. Color is particularly important because it helps the reader follow a category or flow across different dimensions.

Scenario

A parallel sets diagram connects **three categorical variables**, but several unrelated categories use the same color.

a) Critique the Design Flaw

Using the same color for unrelated categories makes it difficult to trace an individual ribbon through the diagram.

The reader cannot determine whether similarly colored ribbons represent the same category or different categories. Therefore, the relationships between the three variables become confusing.

b) Redesign

Assign a **unique and distinguishable color to each category**.

The same category should maintain the **same color throughout the entire diagram**. A categorical color palette with enough distinct hues should be used.

This allows the reader to follow a ribbon from its starting category through the intermediate variable to its final category.

Evidence

When unrelated categories have the same color, color can no longer be used to distinguish their flows.

Interpretation

A consistent color mapping makes the categorical relationships easier to follow and reduces visual confusion.

R Code – Parallel Sets Style Visualization

For R, `ggalluvial` is commonly used to create parallel sets/alluvial diagrams.

```
R
install.packages("ggalluvial")
library(ggplot2)
library(ggalluvial)

data <- data.frame(
  Category1 = c(
    "A", "A", "B", "B",
    "C", "C"
  ),
  Category2 = c(
    "X", "Y", "X", "Z",
    "Y", "Z"
  ),
  Category3 = c(
    "P", "Q", "P", "R",
    "Q", "R"
  ),
  Frequency = c(
    30, 20, 25, 15,
    20, 10
  )
)

ggplot(
  data,
  aes(
    axis1 = Category1,
    axis2 = Category2,
    axis3 = Category3,
    y = Frequency
  )
) +
  geom_alluvium(
    aes(fill = Category1),
    width = 1/12
  ) +
  geom_stratum(
```

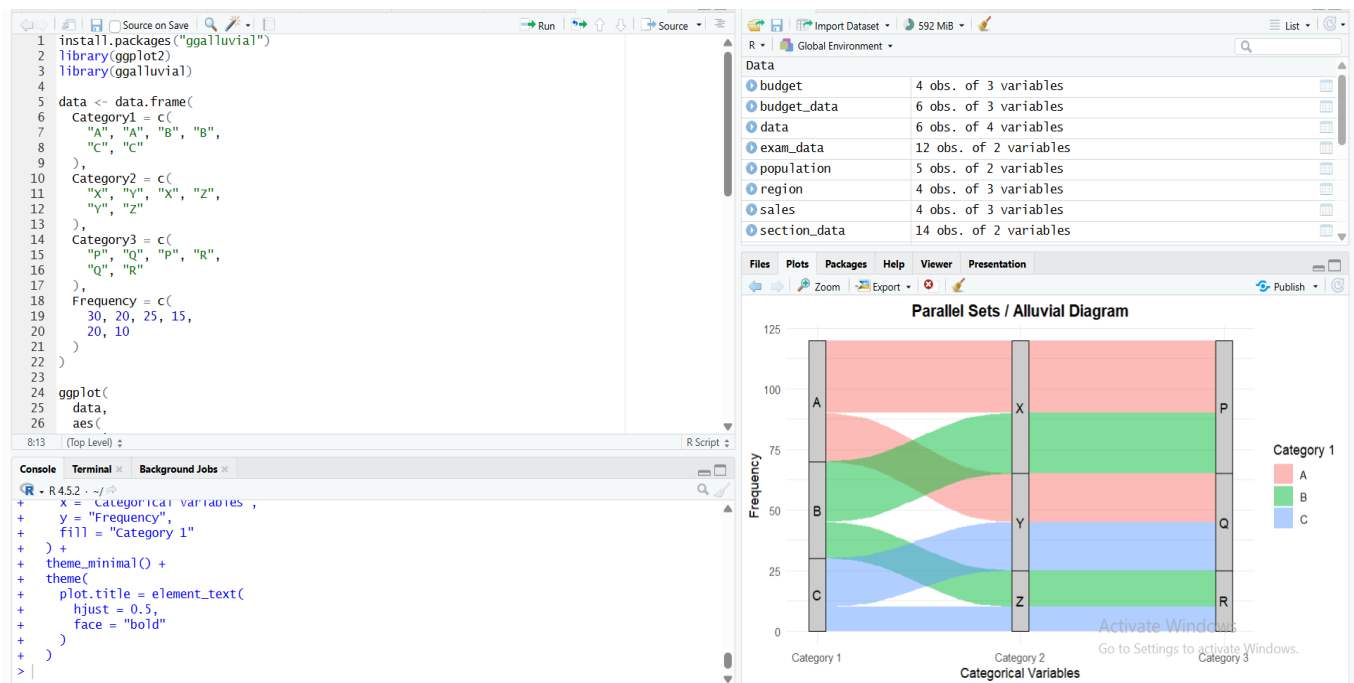
```

width = 1/12,
fill = "grey80",
color = "black"
) +
geom_text(
  stat = "stratum",
  aes(label = after_stat(stratum))
) +
scale_x_discrete(
  limits = c(
    "Category 1",
    "Category 2",
    "Category 3"
  ),
  expand = c(.05, .05)
) +
labs(
  title = "Parallel Sets / Alluvial Diagram",
  x = "Categorical Variables",
  y = "Frequency",
  fill = "Category 1"
) +
theme_minimal() +
theme(
  plot.title = element_text(
    hjust = 0.5,
    face = "bold"
  )
)
)

```

Expected Result

Each category receives a distinct color, and the same category maintains its color throughout the flows.



Conclusion

The original diagram has a **color ambiguity problem**. Assigning unique and consistent colors to categories makes individual flows easier to trace and improves the readability of the parallel sets diagram.